

Sundar Gurumurthy

✉ sundar@sundar.guru • ☎ [\(+44\) 7442278370](tel:+447442278370) • 🌐 [Personal Website](#) •

🌐 [LinkedIn](#) • 🐙 [GitHub](#)

Professional Summary

Simulation engineer specialising in **Python/C++ development**, **FEA**, **physics-based modelling**, **data-driven simulation**, and **statistical analysis** for complex manufacturing and structural processes. Experience includes **process modelling**, **multi-objective optimisation**, **DOE**, **MVEE analysis**, and integration of **laser-scan data** with simulation workflows. Proficient in **scikit-learn**, **PyTorch3D**, **pandas**, **polars**, and custom simulation tooling. Practical hands-on experience in **WAAM robots**, **CNC-based scanning operations**, and experimental design. Motivated to apply scientific computing and analytical modelling to **autonomous manufacturing**, **process optimisation**, and **physics-based digital workflows**.

Technical Skills

Simulation & Modelling

- Physics-based modelling (thermal-mechanical, nonlinear, contact)
- Parameterisable first-principles modelling
- Data-driven surrogate models
- FEA: Abaqus, LS-Dyna, ANSA (automation)
- Multi-objective optimisation (parameter sweeps, process windows)

Programming & Data Science

- Python (automation, modelling, ML/statistics)
- C++, Bash/Linux, Rust
- Scikit-learn, PyTorch3D, Pandas, Polars
- DOE, MVEE, regression, correlation, uncertainty analysis
- Data pipelines, structured scientific datasets

Manufacturing & Automation

- WAAM: process modelling, bead geometry analysis, distortion prediction
- 6-axis robot programming, CNC laser scanning automation
- DfM/DfAM for forging, machining (soft → hard), broaching, grinding, polishing, casting reviews

Tools & Infrastructure

- Plotly, Matplotlib, Seaborn, PyVista, VisPy, GNUplot for scientific visualisation
- Siemens NX, CATIA V5, Teamcenter
- GitLab CI/CD, PBS scheduler
- HDF5, basic MariaDB/SQL, structured data formats

Projects & Applied Modelling

Geometry-Linked Manufacturing Acceptance Criteria for Stress Performance

Cranfield University

2025 – Present

- Developed **geometry-based acceptance limits** linking surface profile metrics to tensile and distortion behaviour.
- Integrated **simulation**, **laser scanning**, and **tensile testing** into lightweight validation workflows.
- Reduced dependency on full physics simulations by building a **fast, data-driven criteria framework**.

Python Library for WAAM & Scan-Based Qualification

Cranfield University

2025 – Present

- Built tools for **laser-scan processing**, **MVEE analysis**, **deviation mapping**, and geometric feature extraction using ML and conventional methods.
- Automated entire process from **scan acquisition** → **data processing** → **statistical analysis** → **reporting**.
- Implemented **interactive 2D/3D dashboards** (Plotly) for manufacturing engineering reviews.
- Used by project partners for **process qualification**, **defect assessment**, and **data-driven manufacturability decisions**.

Process-Driven WAAM Simulation & Optimisation

Cranfield University

2024 – Present

- Applied **thermal/mechanical modelling** to identify optimal deposition parameters.
- Designed **DOE-based sensitivity studies** to analyse process–response behaviour.
- Provided modelling guidance for aerospace and energy components.

Work Experience

Research Assistant

Cranfield University, UK

Jun 2024 – Present

- Built **Python/C++ simulation automation pipelines** for thermal-mechanical modelling and process-window analysis.
- Developed **phase-transformation strain models** using test-driven statistical fitting and thermal-expansion datasets.
- Integrated **laser-scan data** with simulation workflows for geometric conformity assessment and deviation-driven modelling.
- Implemented **rapid reduced-order approaches** producing industry-usable simulation results in minutes.
- Developed **FORTTRAN heat-source models** and **C++ Abaqus subroutines** for nonstandard material/process cases.
- Delivered **technical reports, experimental analyses**, and multi-stakeholder design reviews.

Graduate Engineering Trainee

Sona Comstar, India

Jul 2021 – Jul 2022

- Performed **fatigue, structural and tolerance assessments** for drivetrain and differential systems.
- Conducted **forging DfM**, and manufacturing reviews for **machining (soft → hard machining), broaching, grinding, polishing, and casting**.
- Defined **operational load cases** from torque maps and test data.
- Automated **CAD→FEA pipelines** using Python/VBA, improving turnaround.
- Delivered **stress reports, DFMEA documentation** and supplier handover packages.

Student Trainee – Crash Structures

Mercedes-Benz R&D India

Feb 2021 – Jun 2021

- Built **LS-Dyna models** for tyre, rim and composite crash scenarios.
- Worked on **Lagrangian contact optimisation** for fibre-reinforced rubber composites.
- Automated **ANSA meshing** using Python scripts.
- Supported **test–simulation correlation** for modal/impact behaviour.

Education

MSc by Research, Manufacturing

Cranfield University

Jan 2023 – Oct 2024

Thesis: Improving the Inherent Strain Method for WAAM Simulation

- Reduced distortion error **14% → 3%** via algorithmic optimisation.
- Developed **fast inherent-strain simulation** reducing runtime from days to minutes.

B.E. Mechanical Engineering

BITS Pilani

Aug 2017 – Jun 2021

- CGPA: 7.71/10 (First Class)

Awards

- AIAA/USU SmallSat Travel Award (Blue Origin Sponsored)